Alex Tyh Wang

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Summary

Computer science undergraduate building **GenAI applications and agentic systems**: production LLM agents, tool calling, context engineering, **RAG**-adjacent retrieval patterns, and evaluation harnesses in **Python** and TypeScript.

Education

Rochester Institute of Technology — Rochester, NY
Bachelor of Science in Computer Science — Expected May 2028

Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, C#, SQL
Frameworks and Tools: React, Angular, Next.js, Spring Boot, REST, ASP.NET, Node.js, PostgreSQL, Azure DevOps, Jenkins
AI and Systems: GenAI apps, AI agents, multi-agent systems, subagent architectures, MCP, tool calling, context engineering, RAG, LLM evaluation, model reliability, token optimization, OpenAI API, OpenRouter

Experience

Software Engineer Intern — Jan 2026 – Aug 2026
Paychex — *Rochester, NY*

- Led development of an internal AI developer platform on the **AI Integrations** team that integrated **AI agents and MCP tooling** into enterprise developer workflows; presented the platform to executives.
- Built **LLM evaluation** systems for real developer workflows, comparing model behavior, reliability, and usefulness.
- Reduced token usage by **55%** with a **subagent architecture** while improving internal agent workflow performance.

Student Researcher — Jan 2026 – May 2026
Rochester Institute of Technology — *Rochester, NY*

- Conducted research on large language models and memory.

Software Engineer Intern — Feb 2025 – Aug 2025
Real Wave Capital — *Remote*

- Architected production Angular and TypeScript UIs for autonomous AI agents and rewrote **MCP tool-calling and context strategy**, supporting 100+ daily users and cutting token spend by roughly 50%.
- Integrated OpenAI API and OpenRouter into production agent flows with .NET backend contributions, improving endpoint response times by 25% and scaling to 6,000+ weekly requests.

Projects

Chatio Agentic Behavior Benchmark and Chat App — Jun 2025 – Present
Python, TypeScript, LLM Evaluation, Tool Calling — *chatio.dev*

- Built a roughly 500-prompt LLM benchmark evaluating real-world assistant utility, with ensemble LLM-as-judge scoring and deterministic **Python** validation.
- Created production agent workflows with SSE streaming, **tool calling**, multi-step orchestration, and dynamic parameter optimization.
- Engineered a high-performance persistence layer combining IndexedDB with Cloudflare R2, reducing image storage by 95% through intelligent compression.

Leadership and Activities

Founder and President — Jan 2022 – Present
Bay Area Stem Academy — *San Jose, CA | bayareastemacademy.org*

- Founded a 501(c)(3) nonprofit offering free robotics, machine learning, web development, 3D modeling, and engineering workshops; taught **660+ students** through **15+ free camps** and partnered with **5+ schools**.
- Led website development, curriculum, workshops, mentoring, outreach, and logistics, providing **$35,000+** in free STEM education and donating **$5,000+** to partner organizations and teams.
- Led a collaboration with **SanDisk** and co-hosted a volunteer day at SanDisk's Milpitas campus, mobilizing employees to build STEM kits for future programs.

Event Coordinator and Executive Board Member — Sep 2024 – Present
RIT Asian Culture Society — *Rochester, NY*

- Co-leading a community of **200+ members** and organizing meetings, fundraising, and Night Market, an annual university-wide event with **600 attendees**.